Oliver Yerbury-Hodgson

Application for MPhil in Machine Learning and Machine Intelligence, Cambridge University, UK

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Formal Education

BSc Hons in Discrete Mathematics

University of Warwick

September 2014 - July 2017   Warwick, UK

- Completed a dissertation in the use of structured prediction and classifier chains on missing information in personal calendar data.
- Achieved the top performance deep learning model in the cohort in a Machine Learning module to classify images of fish.
- High performance (89%) in a module exploring BSP (Bulk Synchronous Parallelism) and parallel algorithm complexity.
- Courses included Database Systems, Artificial Intelligence, Machine Learning, Compiler Design, Algorithmic Graph Theory, Advanced Linear Algebra, Combinatorics, and Computer Graphics.

Java | Haskell | Python | Vowpal Wabbit | Event Scheduling

A Levels / GCSEs

Royal Wootton Bassett Academy

September 2010 - July 2014   Royal Wootton Bassett, UK

- 3 A-Levels (A* in Further Mathematics, A* in Mathematics, A in Physics)
- 2 AS-Levels (A in Graphics, B in AS Science)
- Advanced Extension Award in Mathematics
- 12 GCSEs (7 GCSEs at grade A*, 5 GCSEs at Grade A)

Work Experience

Commoditisation and Delivery Engineer

Credit Suisse

September 2017 - Current   London, UK

- Created a Machine Learning and Data Analytics Platform that supports 10+ internal team projects. This included configuration of NVIDIA GPUs and high level architecture design.
- Performed large-scale experimentation through distributed computing (Hadoop) and time-series modelling to detect anomalies in pricing data.
- Built a test suite for a low latency trading system Smart Order Router against market-like conditions.
- Facilitated the surveillance of unstructured data such as electronic communications by using ElasticSearch and a React based UI to satisfy a regulatory requirement.

NVIDIA | Forecasting | Ensemble Models | Hadoop | C++

Technology Intern

Credit Suisse

10 Weeks in July-September 2016   London, UK

- Used entropy to perform dimensionality reduction of trade data, securing one of 10/35 places to convert to full-time employment.

Python | Information Theory

Research Interests

- Technology of Sustainability
- Natural Language Processing
- Computer Vision
- Simulation / Modelling
- Data Visualisation / Data Journalism

Extracurricular

Fundraising

Meningitis Research Foundation / St. Giles Trust

2015 / 2019

- I've fundraised £3750 in total by climbing the Balkan Peaks and Mount Kilimanjaro on separate occasions.

Awards

Best Overall Graduating BSc Student in Discrete Mathematics for 2017

CompSci Department, Warwick University

June 2017

Warwick, UK

- Departmental prize for achieving the highest overall mark of 81.5% in my graduating class.

Discrete Mathematics Third-Year Project Prize for 2017

CompSci Department, Warwick University

June 2017

Warwick, UK

- For the completion of the Discrete Mathematics degree we had to complete a presentation and dissertation of a chosen topic, contributing to 15% of final grade.
- Awarded the departmental prize for the best third year project amongst my class with a mark of 84%.

Departmental Prize for best Data Structures module coursework

Morgan Stanley Sponsored Award

June 2015

Warwick, UK

- In the first year we were tasked with creating a Twitter clone in Java. I was awarded a prize for the best coursework.